

01_data_cleaning.py

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1  """
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3
4  Health Insurance Cost & Claims Analytics
5  Data Cleaning & Preparation
6
7  Extracted directly from the project's analysis notebook (analysis.html).
8
9
10 Run this file first. It expects data.csv in the working directory
11 and produces the cleaned `df` used by 02_eda.py and
12 03_feature_engineering_modeling.py in the same session/kernel.
13 """
14
15
16
17 # Imports
18
19
20 import numpy as np
21 import pandas as pd
22 import matplotlib.pyplot as plt
23 import seaborn as sns
24 import joblib
25
26
27
28
29 # Load data
30
31
32 df = pd.read_csv("data.csv")
33
34 # Duplicate check
35
36
37 n_duplicates = df.duplicated().sum()
38 print(f"Duplicate rows found: {n_duplicates}")
39 if n_duplicates > 0:
40     df = df.drop_duplicates()
41     print(f"Duplicates dropped. Rows remaining: {len(df)}")
42
43
44 # Handling missing values
45
46
47 missing_before = df.isna().sum()
```

```
48 total_missing = missing_before.sum()
49 rows_before = len(df)
50
51 print("Missing values by column:")
52 print(missing_before[missing_before > 0])
53 print(f"\nTotal missing cells: {total_missing}")
54
55
56 # Check whether missingness is concentrated in a particular region or
57 # risk group before blindly dropping it – a silent drop can bias the
58 # training population if missingness isn't random.
59
60
61 if total_missing > 0 and "region" in df.columns:
62     rows_with_na = df[df.isna().any(axis=1)]
63     print("\nMissingness by region (rows with at least one NaN):")
64     print(rows_with_na["region"].value_counts())
65
66 df = df.dropna()
67 rows_after = len(df)
68 pct_dropped = 100 * (rows_before - rows_after) / rows_before
69 print(f"\nRows before drop: {rows_before} | after: {rows_after} "
70       f"| dropped: {rows_before - rows_after} ({pct_dropped:.2f}%)")
71
72 if pct_dropped > 5:
73     print("WARNING: more than 5% of rows dropped for missingness – "
74           "confirm this isn't concentrated in one segment before proceeding.")
75
76 assert df.isna().sum().sum() == 0, "Missing values still present after dropna()"
77
78
79
80 # Summary statistics
81
82
83 print(df.describe(include="all"))
84
```